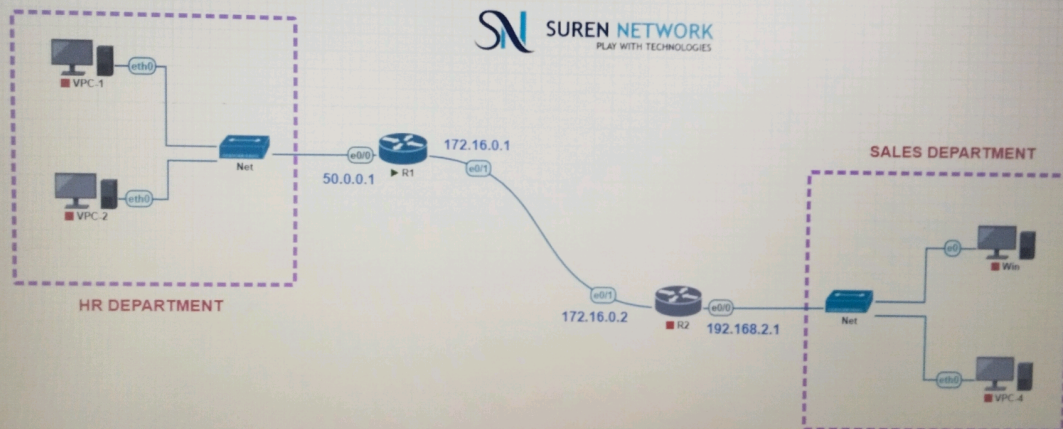




R1

```
R1(config-if)#no sh
R1(config-if)#ex
R1(config)#int e0/1
R1(config-if)#ip add 172.16.0.1 255.255.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.1
%Invalid next hop address (it's this router)
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.2
R1(config)#exit
R1#
```

*Sep 1 13:04:53.733: %SYS-5-CONFIG_I: Configured from console by console

R1#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	50.0.0.1	YES	manual	up	up
Ethernet0/1	172.16.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R1#sh ip route

*Sep 1 13:05:19.824: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

```
50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    50.0.0.0/8 is directly connected, Ethernet0/0
L    50.0.0.1/32 is directly connected, Ethernet0/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/16 is directly connected, Ethernet0/1
L    172.16.0.1/32 is directly connected, Ethernet0/1
S    192.168.2.0/24 [1/0] via 172.16.0.2
```

R1#

R2

```
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#
*Sep 1 13:09:51.481: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Sep 1 13:09:52.481: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config)#int e0/1
Router(config-if)#ip add 172.16.0.2 255.255.0.0
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#
*Sep 1 13:10:29.623: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Sep 1 13:10:30.623: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config)#
Router(config)#ip route 50.0.0.0 255.0.0.0 172.16.0.1
Router(config)#ex
% Ambiguous command: "ex"
Router(config)#exit
Router#
*Sep 1 13:11:41.739: %SYS-5-CONFIG_I: Configured from console by console
Router#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              192.168.2.1     YES manual up          up
Ethernet0/1              172.16.0.2     YES manual up          up
Ethernet0/2              unassigned      YES unset   administratively down down
Ethernet0/3              unassigned      YES unset   administratively down down
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    50.0.0.0/8 [1/0] via 172.16.0.1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/16 is directly connected, Ethernet0/1
L    172.16.0.2/32 is directly connected, Ethernet0/1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, Ethernet0/0
L    192.168.2.1/32 is directly connected, Ethernet0/0
Router#
Router#
```


R1

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks

C 172.16.0.0/16 is directly connected, Ethernet0/1

L 172.16.0.1/32 is directly connected, Ethernet0/1

S 192.168.2.0/24 [1/0] via 172.16.0.2

R1#

R1#

R1#

R1#config t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#ip dhcp pool HR

R1(dhcp-config)#network 50.0.0.0 255.0.0.0

R1(dhcp-config)#default-router 50.0.0.1

R1(dhcp-config)#dns-server 8.8.8.8

R1(dhcp-config)#lease 1 2 3

R1(dhcp-config)#ex

R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50

R1(config)#ex

% Ambiguous command: "ex"

R1(config)#exit

R1#

*Sep 1 13:21:54.764: %SYS-5-CONFIG_I: Configured from console by console

R1#

VPC-1

Welcome to Virtual PC Simulator, version 1.3 (0.8.1)

Dedicated to Daling.

Build time: May 7 2022 15:27:29

Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)

Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)

All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.

Source code and license can be found at vpcs.sf.net.

For more information, please visit wiki.freecode.com.cn.

Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-1> ip dhcp

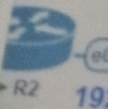
DDORA IP 50.0.0.51/8 GW 50.0.0.1

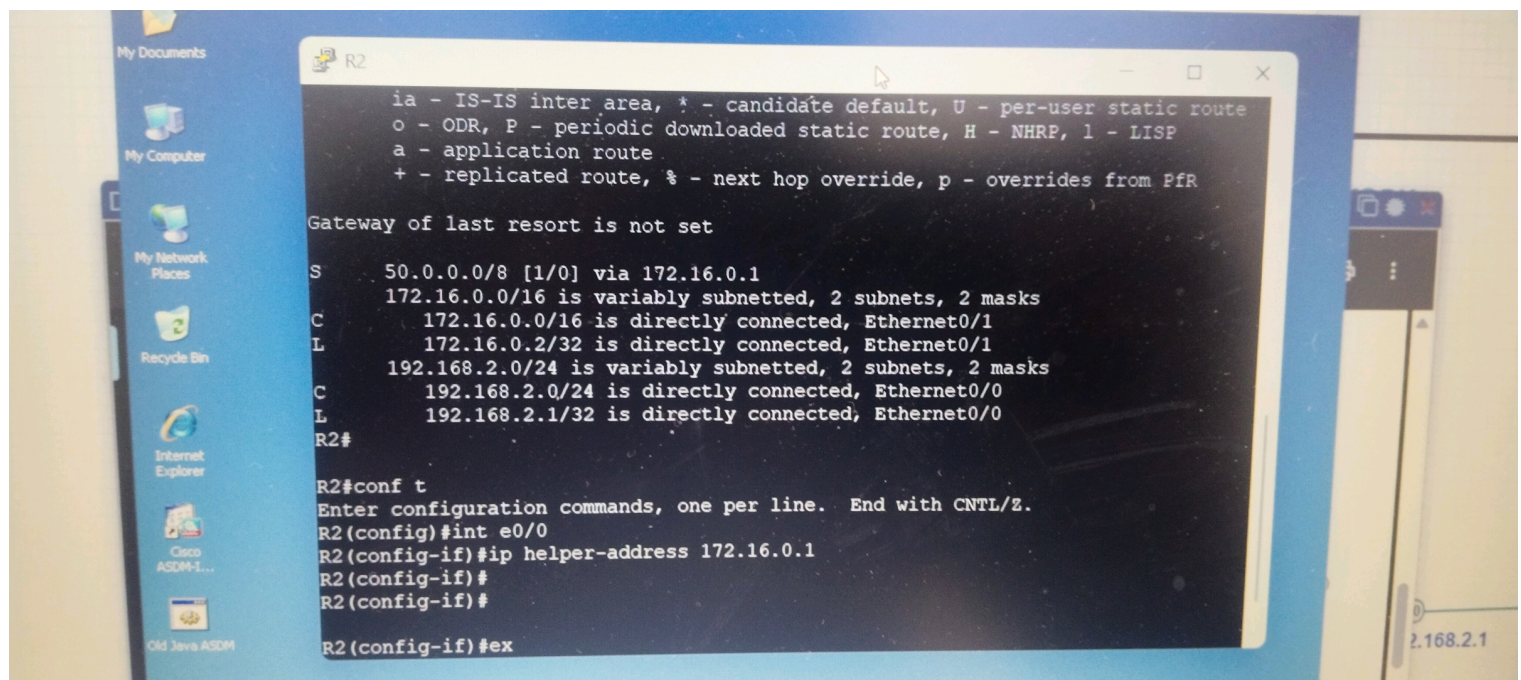
VPC-1> █

R1

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 50.0.0.0 255.0.0.0
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#ex
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50
R1(config)#ex
% Ambiguous command: "ex"
R1(config)#exit
R1#
*Sep  1 13:21:54.764: %SYS-5-CONFIG_I: Configured from console by console
R1#

R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
50.0.0.51       0100.5079.6668.1b  Sep 02 2026 09:00 PM  Automatic
R1#
```





QEMU (Win)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	192.168.114.173
Subnet Mask	:	255.0.0.0
Default Gateway	:	192.168.2.1

C:\Documents and Settings\user>
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	192.168.114.173
Subnet Mask	:	255.0.0.0
Default Gateway	:	192.168.2.1

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	192.168.114.173
Subnet Mask	:	255.0.0.0
Default Gateway	:	192.168.2.1

C:\Documents and Settings\user>